

Homework Assignment 4: Studention Transport Monte Carlo Simulation

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Introduction

This report details the results of an explicit Monte Carlo simulation of Studention transport in a sphere of Exercisium. Exercisium is a fissile material with atomic weight $A = 300$, density $\rho = 30$ g/cm³, and a microscopic cross section $\Sigma_t = 10$ barn.

The simulation advanced active Studentions generation-by-generation. For each Studention, the distance to the next event is determined by $s = -\ln(\xi)/\Sigma_{macro}$. If the advanced particle leaves the sphere radius R , it is deleted. Otherwise, the type of collision is determined probabilistically:

- **Absorption** ($P_0 = 20\%$): The Studention disappears.
- **Scattering** ($P_1 = 50\%$): The Studention scatters in a random 3D direction.
- **Fission** ($P_2 = 30\%$): The original is absorbed, and two new Studentions are emitted in independent random directions.

A sphere is considered critical if at least one of 100 histories starting at the origin becomes critical (surpassing 10,000 active particles). Otherwise, the sphere is noncritical.

Question 1: Critical Radius of the Base Case

Running the bisection search with $N = 100$ histories per check yielded:

- **Critical Radius:** 8.5417 ± 0.2084 cm
- **Critical Mass:** 78.3136 ± 0.1907 kg

Question 2: Critical Mass vs. Density

The simulation was executed for 11 densities linearly spaced between 10 g/cm^3 and 100 g/cm^3 . The resulting critical radius and critical mass are plotted on a logarithmic scale in Figure 1.

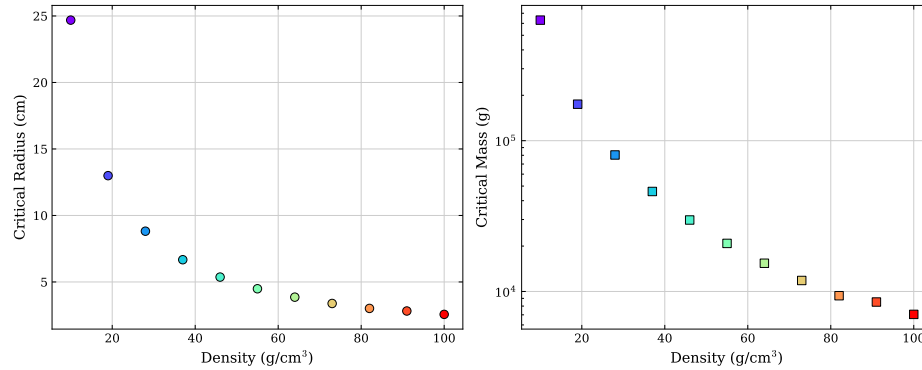


Figure 1: Critical radius (cm) and critical mass (g) as a function of Exercisium density.

Discussion

As the density increases, the mfp (mean free path) decreases, making it less likely for Studentions to escape the sphere at a certain radius. This results in a smaller critical radius, and it is the cause of the decreasing fashion of the graphs in Figure 1.

Question 3: Probability Grid & Parameterization

The critical radius and critical mass were evaluated across 13 different collision probability configurations. The results are plotted against the multiplication parameter $c = P_1 + 2P_2$ in Figure 2.

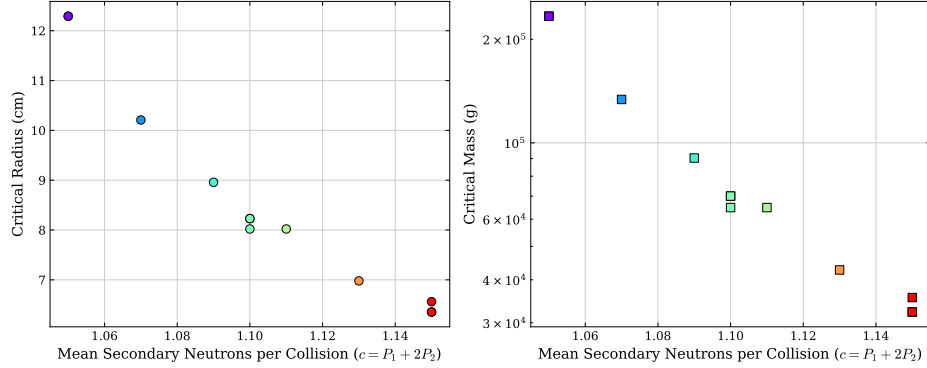


Figure 2: Critical radius (cm) and critical mass (g) as a function of the multiplication parameter $c = P_1 + 2P_2$.

Discussion

In figure 2, we can notice that the critical mass is uniquely determined by the average secondary Studention count c and is not affected by the separate values of P_0, P_1 , and P_2 . As explained in the previous discussion, the shorter the mfp is, the smaller the critical radius is - because the Studention is less likely to escape the sphere at a certain radius. In addition, the more secondary Studentions are produced, the more likely it is that at least one of them will not escape the sphere at a certain radius, so the critical radius is also smaller. This is the decreasing fashion of the graphs in Figure 2.

Question 4: Deep Probability Sweep

A wider grid of 33 different probability configurations was simulated to analyze the critical radius and mass across a broad range of c . The results are shown in Figure 3.

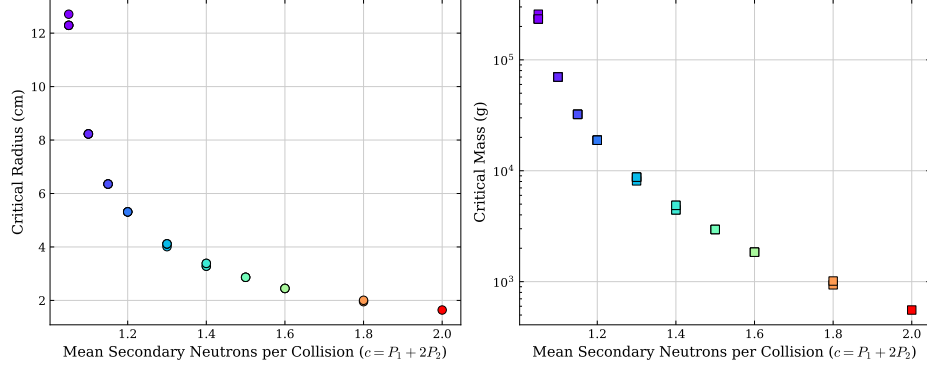


Figure 3: Critical radius (cm) and critical mass (g) for 33 configurations as a function of the multiplication parameter $c = P_1 + 2P_2$.

Discussion

As explained in the previous discussion, the critical mass is uniquely determined by the average secondary Studention count c and is not affected by the separate values of P_0 , P_1 , and P_2 . The entire game is about the multiplication parameter c and the mfp.

Question 5: Comparison with Analytical Approximations

The Monte Carlo radii of Questions 3 and 4 are compared against 2 closed-form results for a bare multiplying sphere (with $c > 1$), both follow from the extrapolated-zero condition, in which the fundamental diffusion mode $\phi \propto \sin(Br)/r$ is driven to zero one extrapolation distance z_0 beyond the surface,

$$\Sigma_t R_c = \frac{\pi}{B/\Sigma_t} - \Sigma_t z_0, \quad (1)$$

so that the two differ only in the pair (B, z_0) . Lengths below are in mean free paths, and are converted to centimetres with $\text{mfp} = 1/\Sigma_t = 1.6667 \text{ cm}$ at $\rho = 30 \text{ g/cm}^3$.

Classical diffusion is defined through the equation:

$$-D\nabla^2\phi + (1-c)\phi = 0 \implies \nabla^2\phi + B^2\phi = 0, \quad B^2 = \frac{c-1}{D} = 3(c-1) \quad (2)$$

and closes the surface with the Marshak condition $z_0 = 2D$:

$$\Sigma_t R_c = \frac{\pi}{\sqrt{3(c-1)}} - \frac{2}{3}. \quad (3)$$

Asymptotic Milne diffusion, the “exact” curve and the asymptotic diffusion solution of this assignment, corrects both members of the pair. It replaces D by the asymptotic diffusion coefficient $D_0 = (c-1)|\nu_0|^2/\Sigma_t$, which reproduces the true relaxation length of the infinite medium and gives $B = \Sigma_t/|\nu_0|$, and it replaces the Marshak surface condition by the exact one, the extrapolation distance of the Milne problem $z_0(c)$ tabulated by Case:

$$\Sigma_t R_c = \pi|\nu_0(c)| - \Sigma_t z_0(c). \quad (4)$$

Equations (3) and (4) are Question 3 of Assignment 1, parts (b) and (a), and are evaluated by the same code.

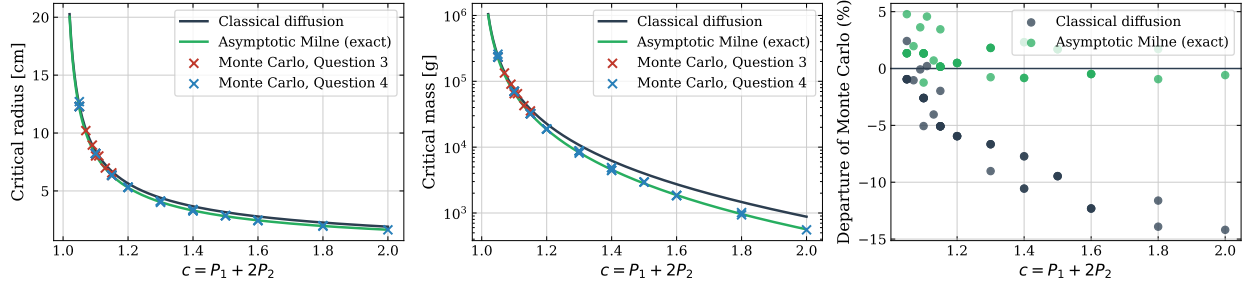


Figure 4: Critical radius and critical mass against c , for the Monte Carlo sweeps of Questions 3 and 4 and the two analytical approximations. The right panel is the signed departure of each Monte Carlo radius from each theory.

Discussion

The asymptotic Milne curve tracks the Monte Carlo over the whole range, staying inside $\pm 2\%$ except at two low- c points, while the classical curve drifts steadily away, from a fraction of a per cent near $c = 1$ to -14% at $c = 2$. The bisection stops at $dr < 0.05r$, so a Monte Carlo radius is

c	R_{MC} [cm]	Departure of Monte Carlo (%)	
		Classical	Asymptotic Milne
1.05	12.361	−0.38	+1.92
1.07	10.208	−1.03	+1.97
1.09	8.958	−0.08	+3.62
1.10	8.199	−2.95	+0.96
1.11	8.021	+0.22	+4.57
1.13	6.979	−4.04	+0.71
1.15	6.384	−4.64	+0.63
1.20	5.313	−5.95	+0.49
1.30	4.089	−7.25	+1.17
1.40	3.333	−9.14	+0.74
1.50	2.865	−9.46	+1.70
1.60	2.448	−12.31	−0.48
1.80	1.979	−12.76	+0.38
2.00	1.641	−14.19	−0.58

Table 1: Mean Monte Carlo critical radius at each distinct c , against the two approximations. Configurations sharing a c are averaged.

itself only determined to about $\pm 2.5\%$. The asymptotic agreement is therefore at the noise floor of the simulation, and the point-to-point scatter in the departure panel of Figure 4 is the stopping rule rather than a failure of the theory.

Classical diffusion is accurate near $c = 1$ because that is the limit in which the P_1 closure it rests on becomes exact. As $c \rightarrow 1$ the medium neither absorbs nor multiplies on average, a Stueckelberg survives many collisions before it is lost, and the angular flux relaxes to near-isotropy between events. Formally, the diffusion relaxation length and the true one converge in that limit: $1/B$ exceeds $|\nu_0|$ by only 2% at $c = 1.05$, because both diverge like $[3(c - 1)]^{-1/2}$ (actually, this was derived in the course as the asymptotic limit of the exact $|\nu_0|$ at $c \rightarrow 1$),

The accuracy falls away as c increases for the same reason in reverse. A strongly multiplying sphere is critical while it is only a couple of mean free paths across so that the leakage is a large fraction of the total loss, and the angular distribution near the surface is strongly forward-peaked. This is where the P_1 closure fails, yet the asymptotic Milne curve remains accurate because it is based on the exact relaxation length.